



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Daniel Stockwell
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Data collection
 - Data wrangling
 - Exploratory Data Analysis with Data Visualisation
 - Exploratory Data Analysis with SQL
 - Building an interactive map with Folium
 - Building a Dashboard with Plotly Dash
- Summary of all results
 - Exploratory Data Analysis results
 - Interactive analytics demo in screenshots
 - Predictive analysis results

Introduction

- Project background and context
 - SpaceX has significantly lowered the cost of orbital launches, primarily through the reusability of its Falcon 9 first-stage boosters. While a typical launch by other providers costs upwards of \$165 million, SpaceX offers launches for \$62 million, a reduction largely due to booster recovery and reuse.
 - Predicting whether a first-stage booster will successfully land is critical to estimating launch cost and operational efficiency. This project applies machine learning to publicly available SpaceX launch data to predict the success of first-stage landings.
- Problems you want to find answers
 - What impact do variables like payload mass, launch site, orbit, and booster history have on landing success.
 - Has the first-stage landing success rate improved over time.
 - Which classification algorithm delivers the highest accuracy when predicting landing outcomes.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Using SpaceX Rest API
 - Using Web Scrapping from Wikipedia
- Perform data wrangling
 - Filtering the data – Dealing with missing values
 - Using One Hot Encoding to prepare the data to a binary classification
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Building, tuning and evaluation of classification models to ensure the best results

Data Collection

Data Collection Process:

- SpaceX REST API – Provided structured technical data.
- Wikipedia Web Scraping – Supplied additional launch context not available via the API.

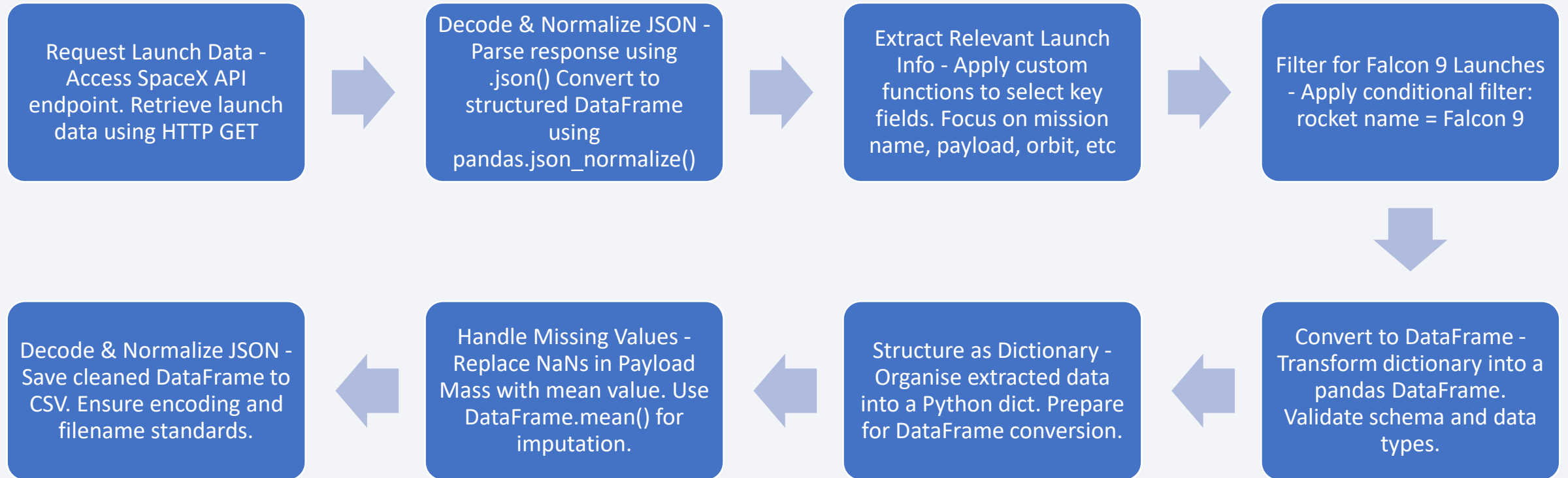
Collected via the SpaceX REST API:

- FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude

Scraped Wikipedia Fields

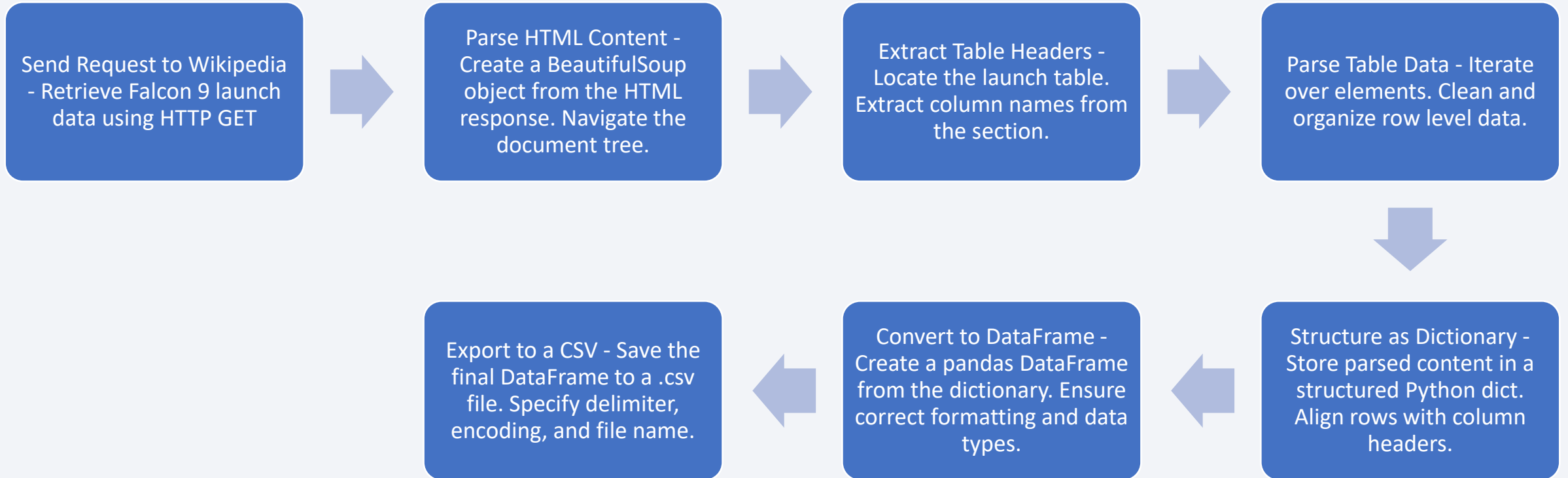
- Extracted from the launch history table on Wikipedia:
- Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

Data Collection – SpaceX API



GitHub URL: Data Collection API

Data Collection - Scraping



GitHub URL: Data Collection with Web Scraping

Data Wrangling

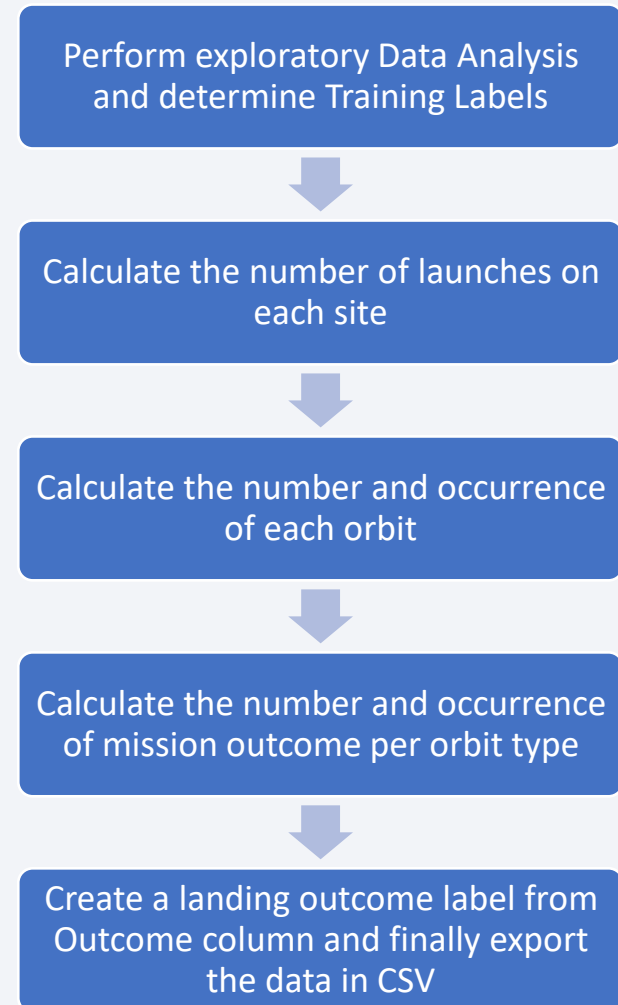
The dataset contains various booster landing outcomes, categorized by method and success:

- True/False Ocean – Landing in the ocean
- True/False RTLS – Ground pad landing (Return to Launch Site)
- True/False ASDS – Drone ship landing (Autonomous Spaceport Drone Ship)

For model training, these outcomes were converted into binary labels:

- 1 → Successful landing
- 0 → Unsuccessful landing

GitHub URL: Data Wrangling



EDA with Data Visualization

- Scatter plots: Examined relationships between variables such as Flight Number, Payload Mass, Launch Site, Orbit Type, and Success Rate to identify potential predictive features.
- Bar charts: Compared success rates across categorical variables like Launch Site and Orbit Type.
- Line charts: Tracked trends in success rates over time.

GitHub URL: EDA with Data Visualisation

EDA with SQL

- Listed unique launch sites. Retrieved 5 launch sites starting with "CCA".
- Calculated total payload by NASA (CRS) launches. Found average payload for booster F9 v1.1.
- Identified first successful ground pad landing date. Listed boosters with successful drone ship landings and payload 4000–6000.
- Counted successful and failed missions.
- Found booster versions with max payload. Retrieved failed drone ship landings in 2015 with booster and launch site. Ranked landing outcome counts between 2010-06-04 and 2017-03-20

GitHub URL: EDA with SQL

Build an Interactive Map with Folium

- Added a marker with a circle, popup, and text label at NASA Johnson Space Center using its latitude and longitude as the initial map location.
- Plotted markers with circles, popups, and text labels for all launch sites, illustrating their geographic locations relative to the Equator and nearby coastlines.
- Used coloured markers clustered by launch outcome green for successful and red for failed launches to highlight launch site success rates.
- Drew coloured lines from KSC LC-39A to nearby landmarks such as railway, highway, coastline, and closest city to visualize proximity distances

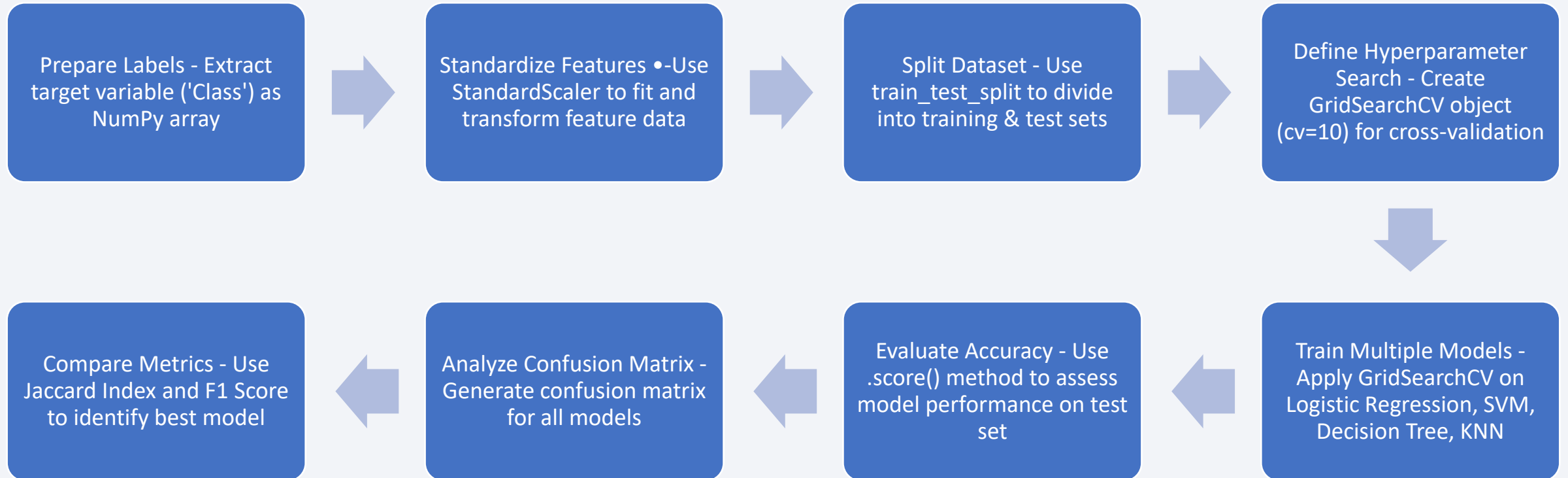
GitHub URL: Interactive Map with Folium

Build a Dashboard with Plotly Dash

- Launch Sites Dropdown: Enables selection of a specific launch site for focused analysis.
- Success Launches Pie Chart: Displays total successful launches for all sites or success vs. failure counts for the selected site.
- Payload Mass Slider: Allows filtering data by a customizable payload mass range.
- Payload vs. Success Scatter Chart: Visualizes the correlation between payload mass and launch success across different booster versions.

GitHub URL: [Dashboard with Plotly Dash](#)

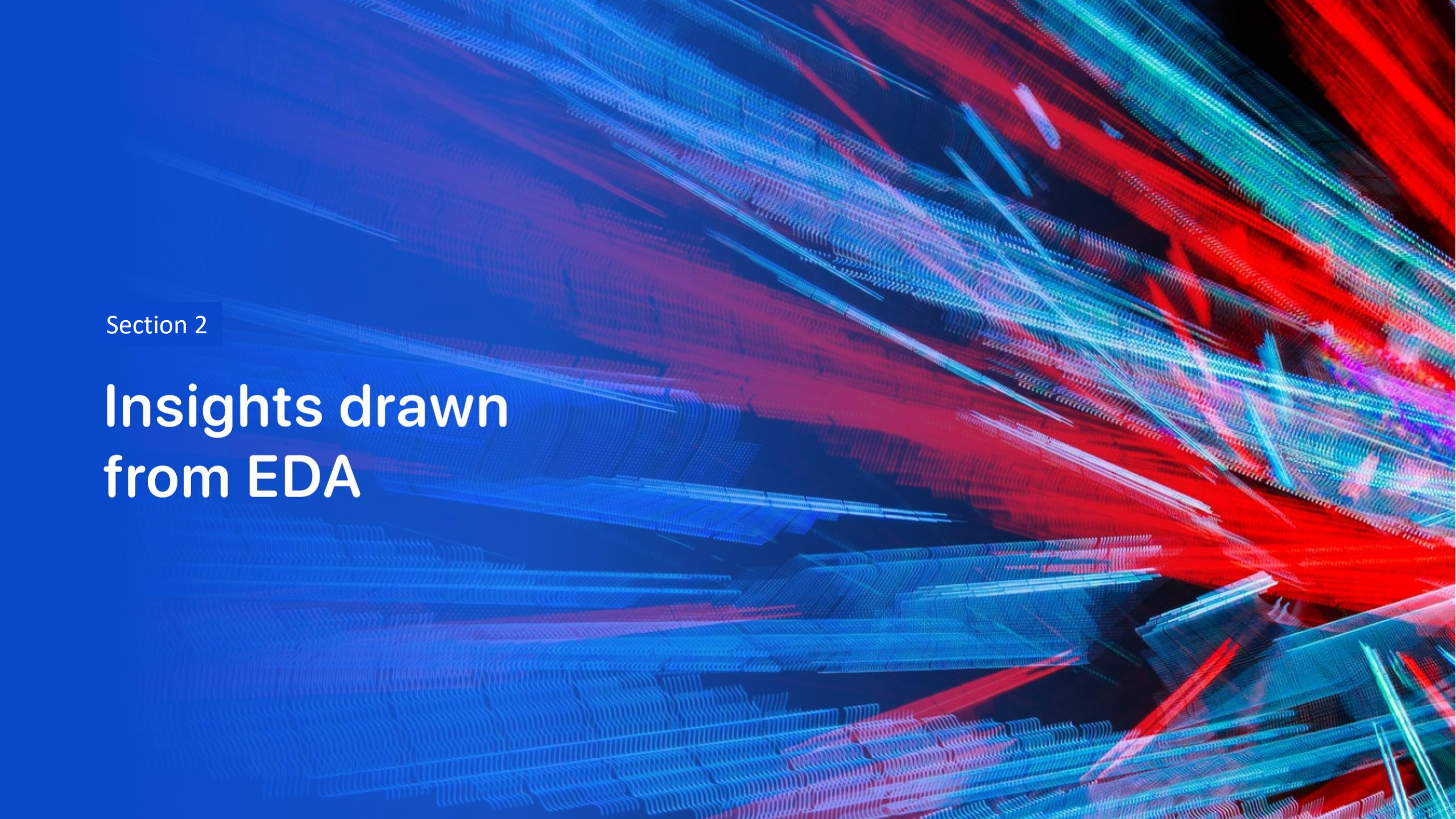
Predictive Analysis (Classification)



GitHub URL: Predictive Analysis (Classification)

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

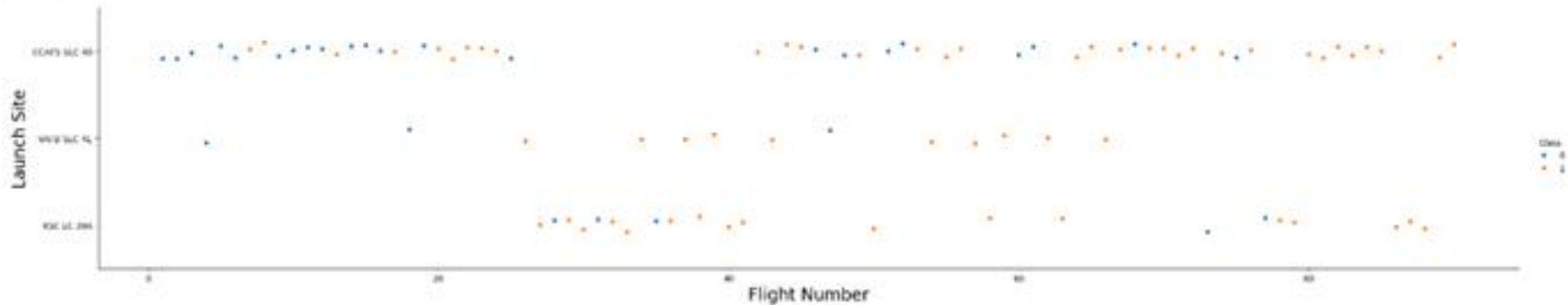
The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

```
[7]: # Plot a scatter point chart with x axis to be Flight Number and y axis to be the Launch site, and hue to be the class value
sns.catplot(x='FlightNumber', y='LaunchSite', hue='Class', data=df, aspect=5)
plt.xlabel('Flight Number', fontsize=20)
plt.ylabel('Launch Site', fontsize=20)
plt.show()
```

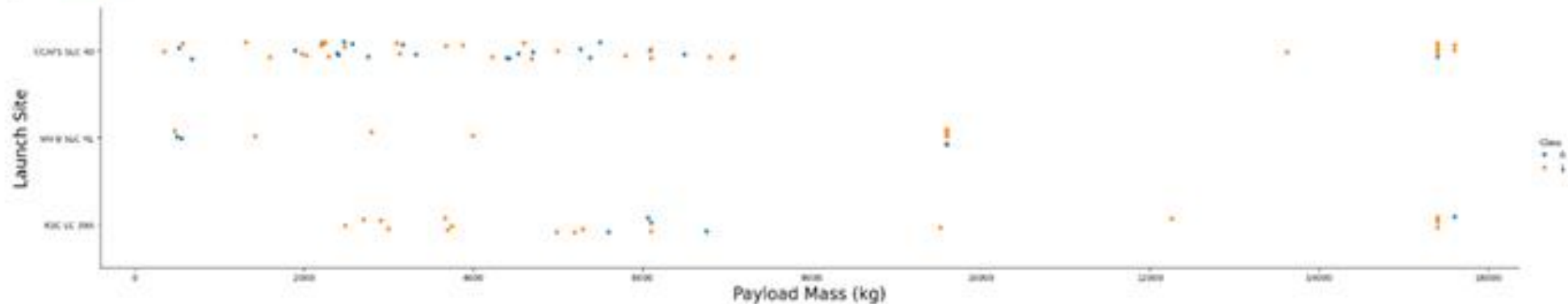


Explanation:

- Early Falcon 9 flights experienced failures, while recent launches have achieved consistent success.
- The CCAFS SLC-40 launch site accounts for approximately 50% of all launches.
- Launch sites VAFB SLC-4E and KSC LC-39A demonstrate higher success rates.
- There is a clear trend indicating increasing success rates with each subsequent launch.

Payload vs. Launch Site

```
[8]: # Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
sns.catplot(x='PayloadMass', y='LaunchSite', hue='Class', data=df, aspect = 5)
plt.xlabel('Payload Mass (kg)', fontsize=20)
plt.ylabel('Launch Site', fontsize=20)
plt.show()
```



Explanation:

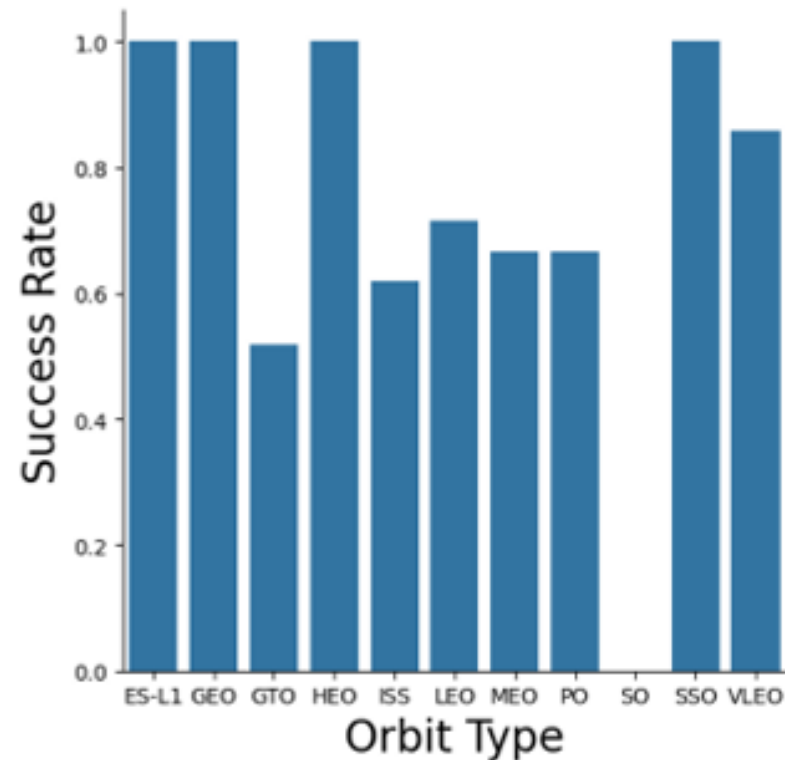
- Across all launch sites, higher payload mass is generally associated with higher success rates.
- Most launches carrying payloads over 7000 kg were successful.
- KSC LC-39A achieved a 100% success rate for launches with payloads under 5500 kg.

Success Rate vs. Orbit Type

Explanation:

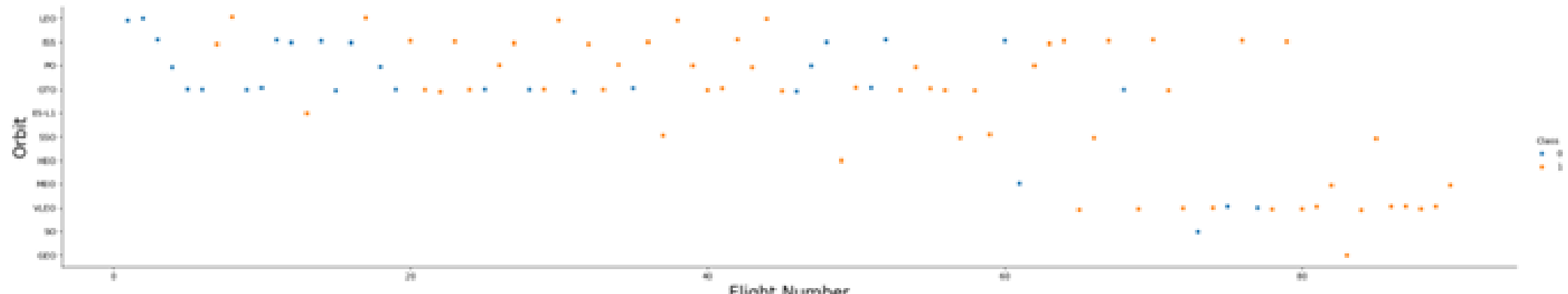
- 100% Success Rate:
 - ES-L1, GEO, HEO, SSO
- 0% Success Rate:
 - SO
- Moderate Success Rate (50%–85%):
 - GTO, ISS, LEO, MEO, PO, VLEO

```
[9]: # HINT use groupby method on Orbit column and get the mean of Class column
sns.catplot(x='Orbit', y='Class', data=df.groupby('Orbit')['Class'].mean().reset_index(), kind='bar')
plt.xlabel('Orbit Type',fontsize=20)
plt.ylabel('Success Rate',fontsize=20)
plt.show()
```



Flight Number vs. Orbit Type

```
[10]: # Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.catplot(x = 'FlightNumber', y = 'Orbit', hue = 'Class', data = df, aspect = 5)
plt.xlabel('Flight Number', fontsize = 20)
plt.ylabel('Orbit', fontsize = 20)
plt.show()
```

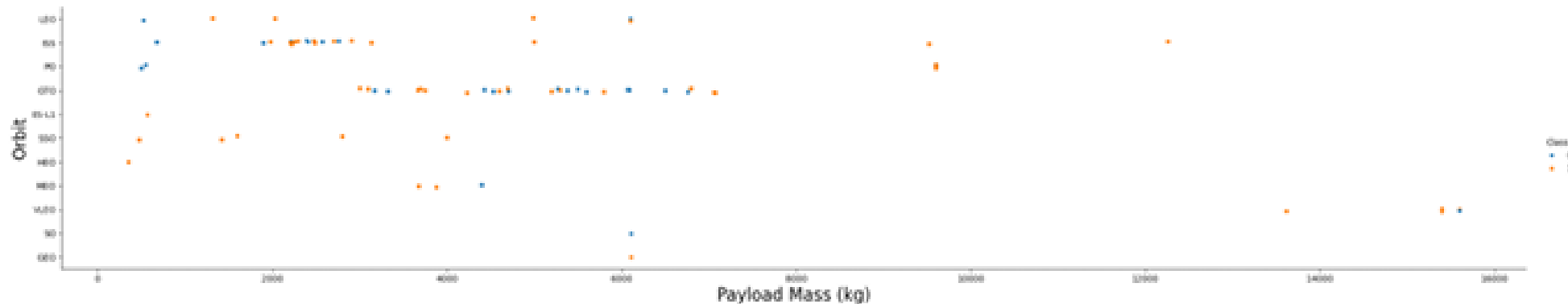


Explanation:

- In LEO (Low Earth Orbit), launch success appears to improve with an increasing number of flights, indicating a positive correlation with experience.
- In contrast, for GTO (Geostationary Transfer Orbit), there is no clear relationship between flight number and success rate.

Payload vs. Orbit Type

```
[11]: # Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
sns.catplot(x = 'PayloadMass', y = 'Orbit', hue = 'Class', data = df, aspect = 5)
plt.xlabel('Payload Mass (kg)', fontsize = 20)
plt.ylabel('Orbit', fontsize = 20)
plt.show()
```



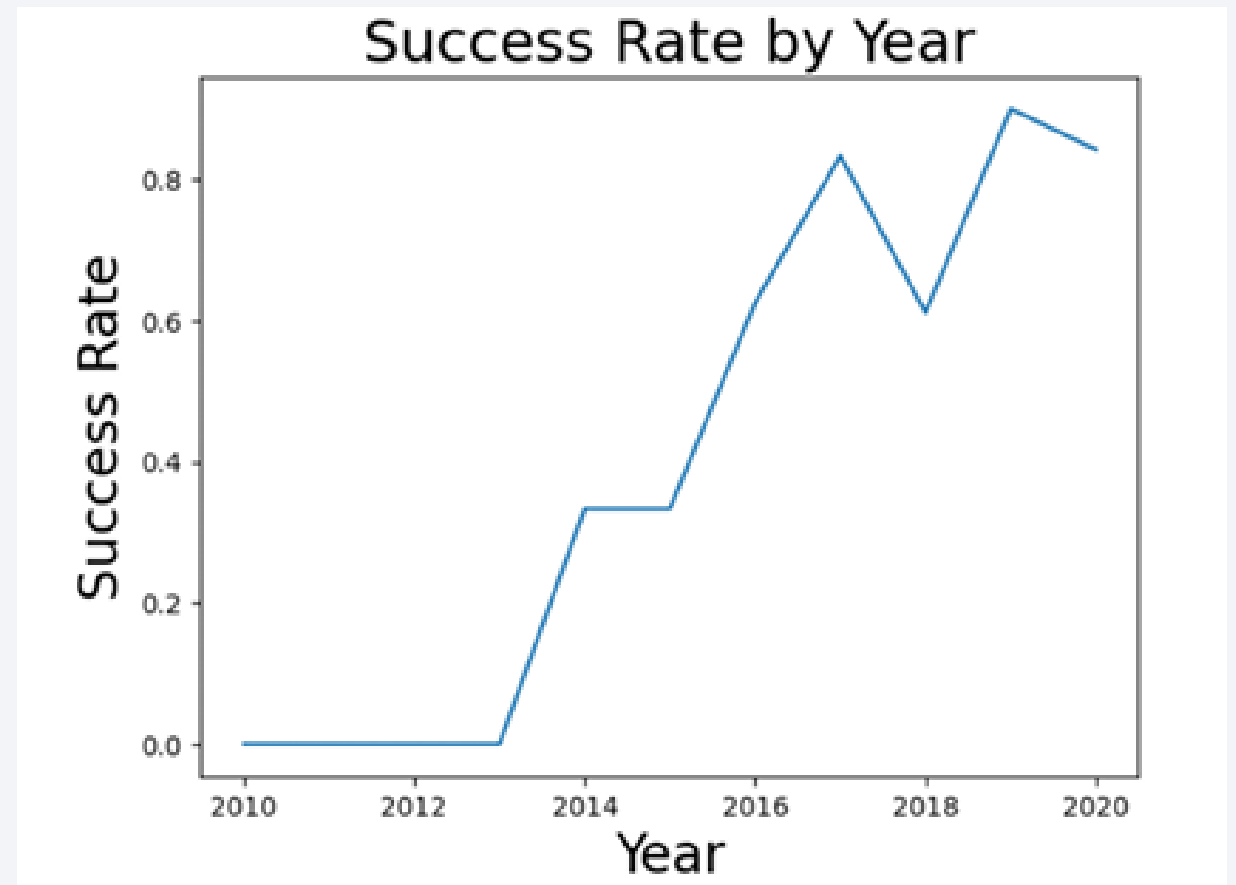
Explanation:

- Heavy payloads negatively impact success rates for GTO orbits.
- Conversely, heavy payloads have a positive effect on success in Polar and LEO (ISS) orbits.

Launch Success Yearly Trend

Explanation:

- The success rate since 2013 kept increasing till 2020



All Launch Site Names

Explanation:

- Displaying the names of the unique launch sites in the space mission

Display the names of the unique launch sites in the space mission

```
[13]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[13]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
[24]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[24]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Explanation:

- Displaying 5 records where launch sites begin with the string 'CCA'.

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[15]: %sql SELECT SUM("Payload_Mass__kg_") AS Total_Payload_Mass FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db  
Done.
```

```
[15]: Total_Payload_Mass  
      45596
```

Explanation:

- Displaying the total payload mass carried by boosters launched by NASA (CRS)

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
[16]: %sql SELECT AVG("Payload_Mass__kg_") AS Avg_Payload_Mass FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[16]: Avg_Payload_Mass
```

```
2928.4
```

Explanation:

- Displaying average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

```
[17]: %sql SELECT MIN("Date") AS First_Successful_Ground_Pad_Landing FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
* sqlite:///my_data1.db
Done.
[17]: First_Successful_Ground_Pad_Landing
      2015-12-22
```

Explanation:

- Displaying average payload mass carried by booster version F9 v1.1

Successful Drone Ship Landing with Payload between 4000 and 6000

Explanation:

- List the names of boosters that have successfully landed on a drone ship and carried payloads between 4000 kg and 6000 kg.

```
[10]: Sql SELECT DISTINCT "Booster_Version" FROM SPACETABLE WHERE "Mission_Outcome" = "Success" AND "PAYLOAD_PASS_KG_" > 4000 AND "PAYLOAD_PASS_KG_" < 6000;
```

* sqlite:///my_data1.db
Done.

```
[10]: Booster_Version
```

FB v1.1
FB v1.1 B1011
FB v1.1 B1014
FB v1.1 B1016
FB FT B1020
FB FT B1022
FB FT B1025
FB FT B1030
FB FT B1021.2
FB FT B1032.1
FB B4 B1040.1
FB FT B1031.2
FB FT B1032.2
FB B4 B1040.2
FB B5 B1046.2
FB B5 B1047.2
FB B5 B1048.3
FB B5 B1051.2
FB B5B1000.1
FB B5 B1058.2
FB B5B1002.1

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
[26]: %sql SELECT "Mission_Outcome", COUNT(*) AS total_number FROM SPACEXTABLE GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db  
Done.
```

```
[26]:
```

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Explanation:

- Listing the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

Explanation:

- Listing the names of the booster versions which have carried the maximum payload mass

```
[21]: %sql SELECT "Booster_Version", "Payload_Mass" FROM SPACEXTABLE WHERE "Payload_Mass" = (SELECT MAX("Payload_Mass") FROM SPACEXTABLE);
```

* sqlite:///my_data1.db
Done.

```
[21]:
```

Booster_Version	Payload_Mass
F9 v1.0 B0003	Payload_Mass
F9 v1.0 B0004	Payload_Mass
F9 v1.0 B0005	Payload_Mass
F9 v1.0 B0006	Payload_Mass
F9 v1.0 B0007	Payload_Mass
F9 v1.1 B1003	Payload_Mass
F9 v1.1	Payload_Mass
F9 v1.1	Payload_Mass
F9 v1.1	Payload_Mass
F9 v1.1	Payload_Mass
F9 v1.1	Payload_Mass
F9 v1.1 B1011	Payload_Mass
F9 v1.1 B1010	Payload_Mass
F9 v1.1 B1012	Payload_Mass
F9 v1.1 B1013	Payload_Mass
F9 v1.1 B1014	Payload_Mass
F9 v1.1 B1015	Payload_Mass

2015 Launch Records

```
[42]: %sql SELECT strftime('%m', Date) AS month_number, strftime('%Y-%m-%d', Date) AS Date, Booster_Version, Launch_Site, Landing_Outcome FROM SPACEXTABLE WHERE Landin
```

▪ sqlite:///my_data1.db
Done.

```
[42]:
```

month_number	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAPS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAPS LC-40	Failure (drone ship)

Explanation:

- Listing the failed landing outcomes in drone ship, their booster versions and launch site names for the months in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
[46]: %sql SELECT Landing_Outcome, COUNT(*) AS Landing_Outcomes FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Lan
```

```
* sqlite:///my_data1.db  
Done.
```

```
[46]:
```

Landing_Outcome	Landing_Outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Explanation:

- Rank landing outcomes (e.g., Failure on drone ship, Success on ground pad) by their count between 2010-06-04 and 2017-03-20, sorted in descending order.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is dark blue with bright yellow and orange lights from cities and towns. The horizon line is visible, separating the dark blue of the atmosphere from the black of space.

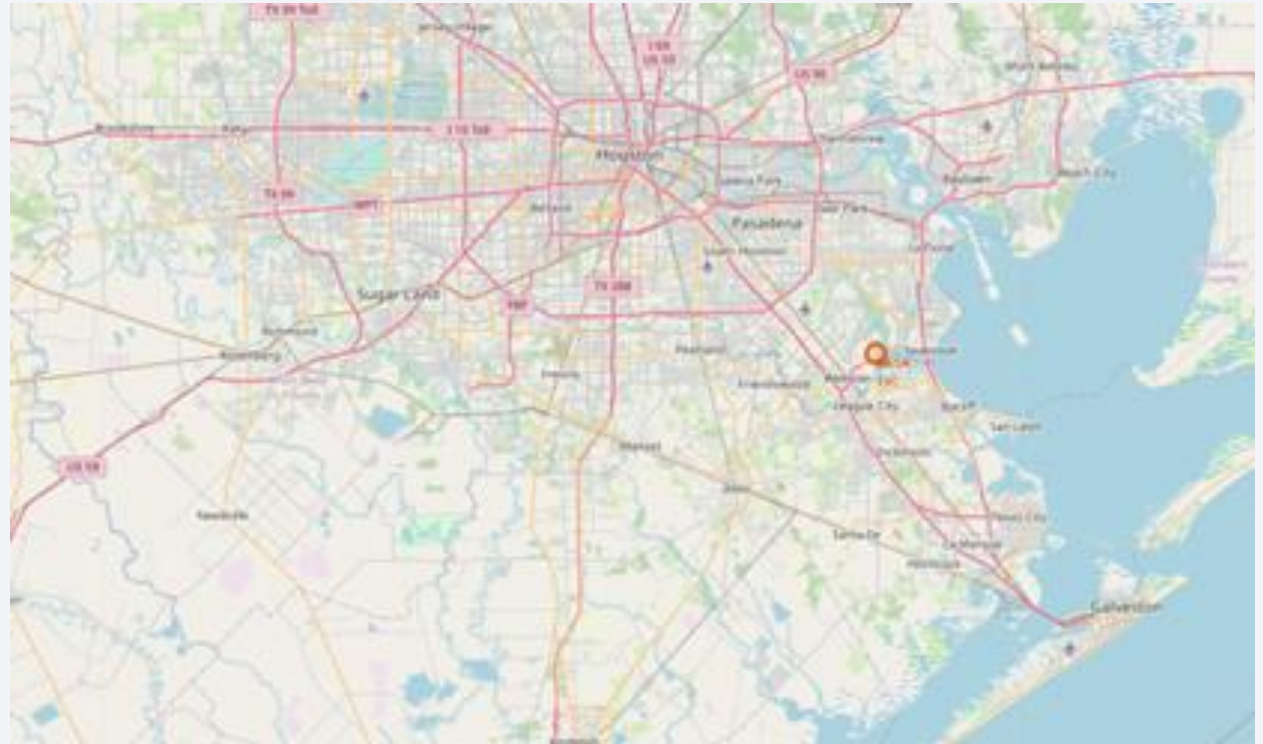
Section 3

Launch Sites Proximities Analysis

Map Screenshot All launch sites' location markers on global map

Explanation:

- Most launch sites are located near the Equator, where the Earth's surface rotates at approximately 1670 km/h—the fastest rotational speed on the planet. Launching from the equator provides rockets with an initial velocity boost due to inertia, aiding in achieving and maintaining orbital speed efficiently.
- Additionally, all launch sites are situated close to the coastline. Launching rockets over the ocean minimizes the risk of falling debris or accidents impacting populated areas, enhancing safety for surrounding communities



Colour-labeled launch record on the map

Explanation:

- Color-coded markers clearly indicate launch site success rates:
 - Green: Successful launch
 - Red: Failed launch
- The launch site KSC LC-39A notably demonstrates a very high success rate

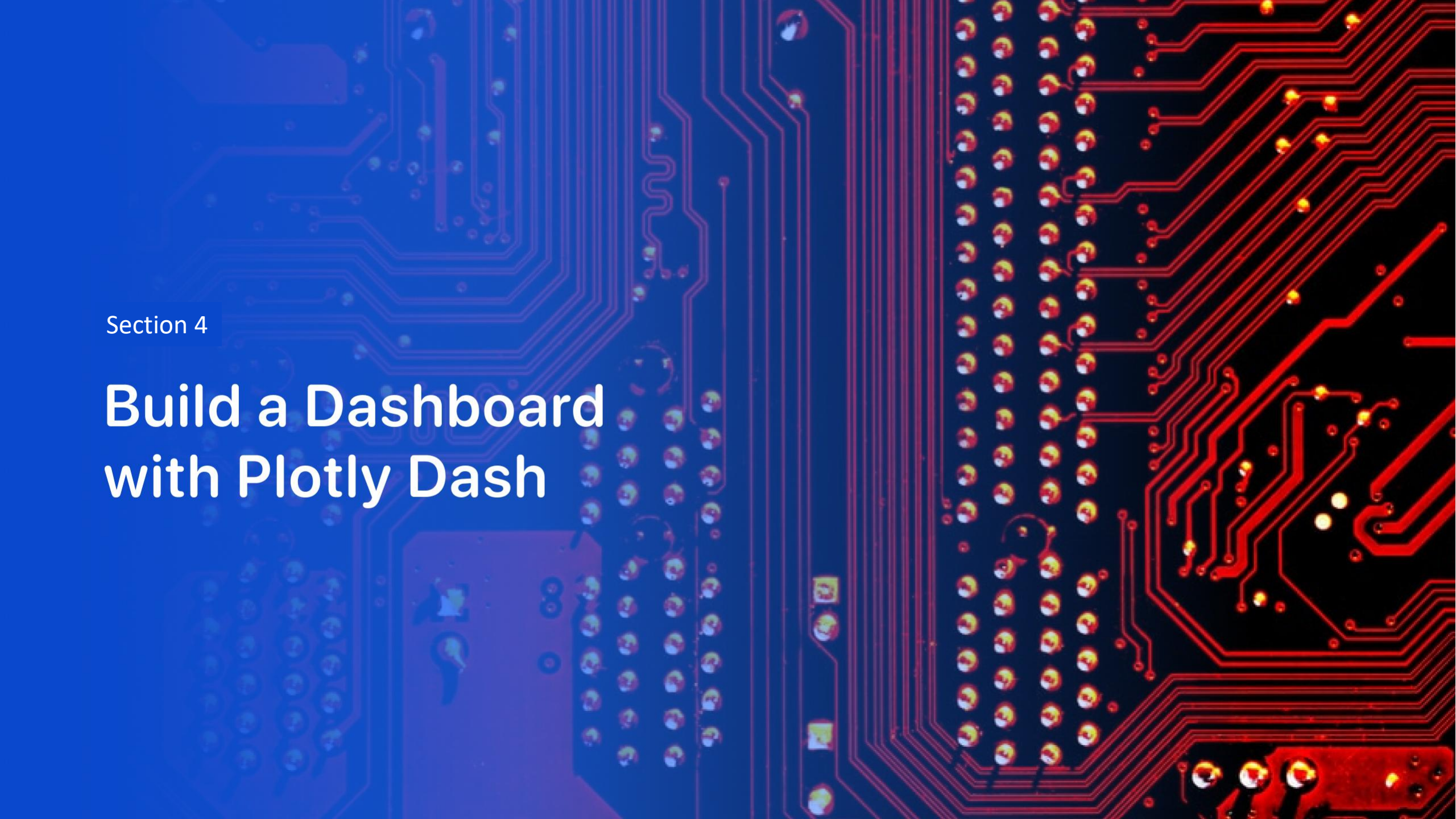


Distance from the launch site KSC LC-39A to its proximities

Explanation:

- Visual analysis shows that KSC LC-39A is located relatively close to key infrastructures and populated areas:
 - Railway: 15.23 km
 - Highway: 20.28 km
 - Coastline: 14.99 km
 - Closest city (Titusville): 6.32 km
- Given that a failed rocket traveling at high speeds can cover 15–20 km within seconds, proximity to these areas presents potential safety risks to nearby populations and infrastructure.





Section 4

Build a Dashboard with Plotly Dash

Launch success count for all sites

Total Success Launches by Site



Explanation:

- The chart clearly shows that from all the sites, KSC LC-39A has the most successful launches

Launch site with highest launch success ratio

Total Success Launches for Site KSC LC-39A

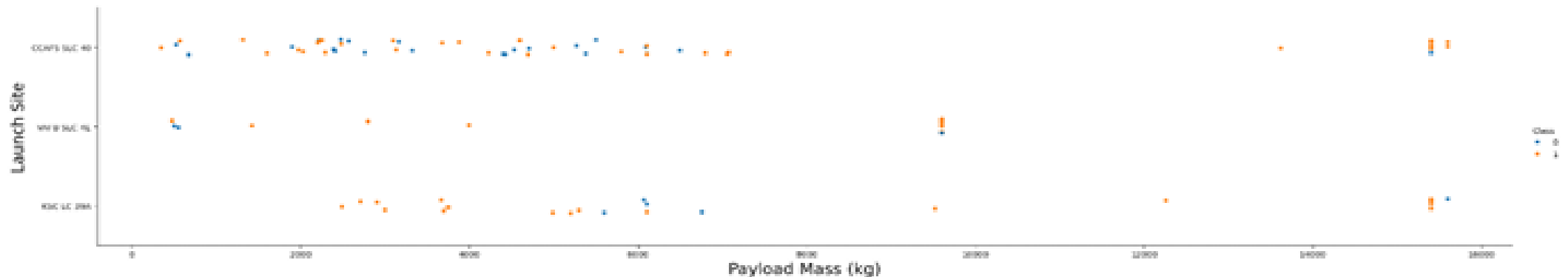


Explanation:

- KSC LC-39A exhibits the highest launch success rate at 76.9%, with 10 successful landings and only 3 failures

Payload Mass Vs. Launch Outcome for all sites

```
[8]: # Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
sns.catplot(x='PayloadMass', y='LaunchSite', hue='Class', data=df, aspect = 5)
plt.xlabel('Payload Mass (kg)', fontsize=20)
plt.ylabel('Launch Site', fontsize=20)
plt.show()
```



Explanation:

- The charts show that payloads between 2000 and 5500 kg have the highest success rate

Section 5

Predictive Analysis (Classification)

Classification Accuracy

```
[58]:
```

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.833333	0.845070	0.833333	0.819444
F1_Score	0.909091	0.916031	0.909091	0.900763
Accuracy	0.866667	0.877778	0.866667	0.855556

Explanation:

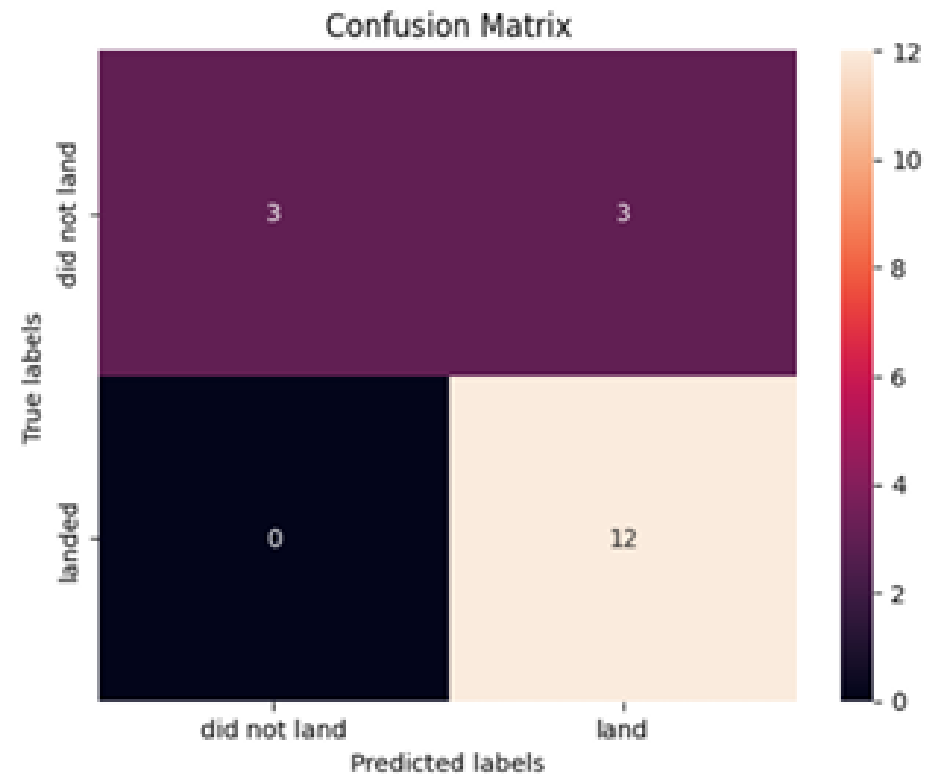
- Evaluation on the full dataset indicates that the SVM model outperforms others, achieving the highest accuracy and superior overall scores.

Confusion Matrix

Explanation:

- The confusion matrix shows that logistic regression effectively distinguishes between classes. However, the primary issue observed is a higher rate of false positives.

```
[19]: yhat=logreg_cv.predict(X_test)  
      plot_confusion_matrix(Y_test,yhat)
```



Conclusions

- The SVM is the best-performing algorithm for this dataset.
- Launches with lower payload mass tend to have higher success rates compared to heavier payloads.
- Most launch sites are located near the Equator and close to the coastline.
- Launch success rates have improved steadily over the years.
- KSC LC-39A records the highest success rate among all launch sites.
- Orbits ES-L1, GEO, HEO, and SSO show a 100% success rate

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

